

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

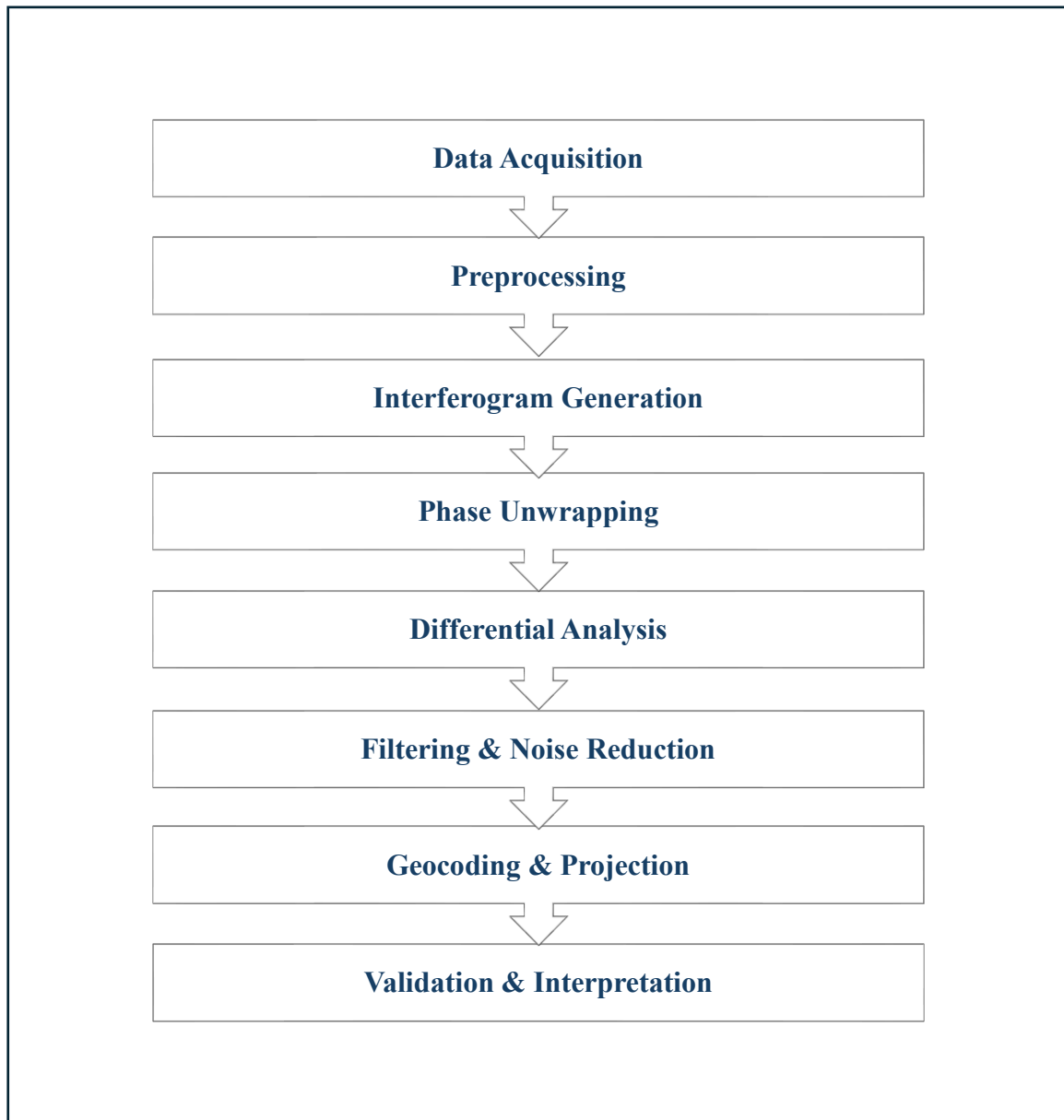
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
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S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 893+270

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34

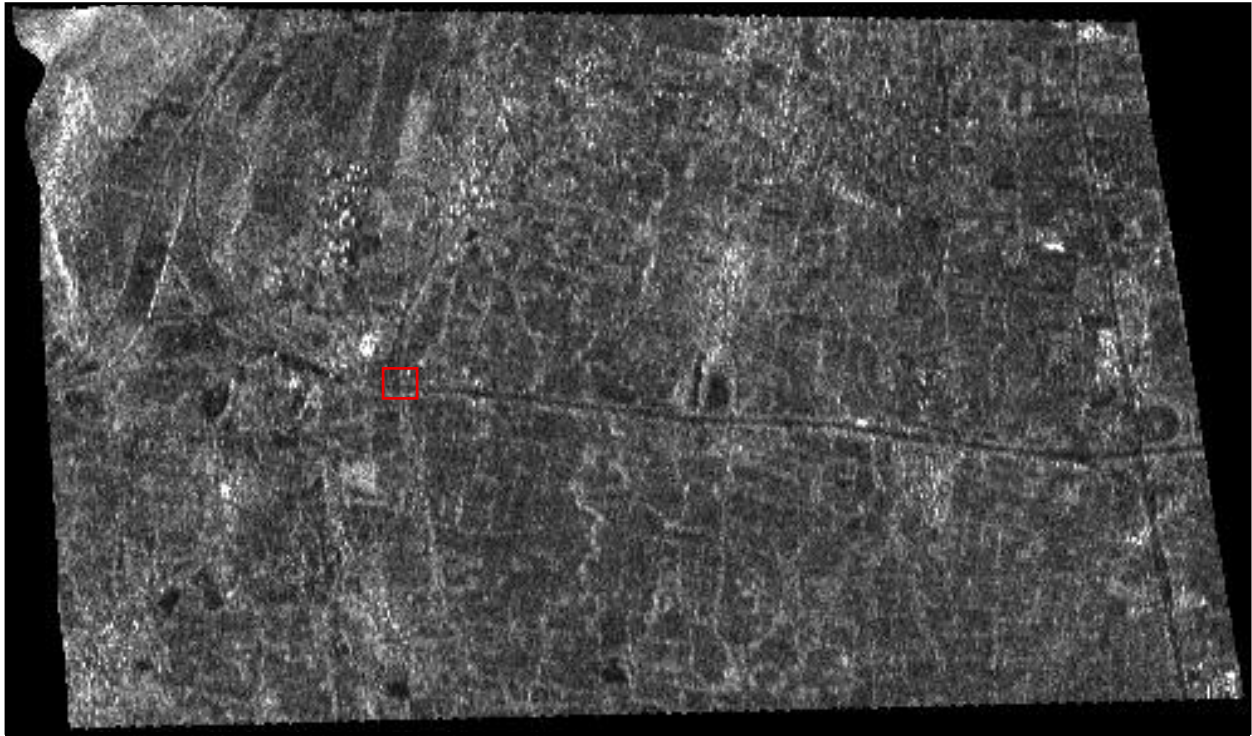
Flight Direction - DESCENDING

Polarization – VV+VH

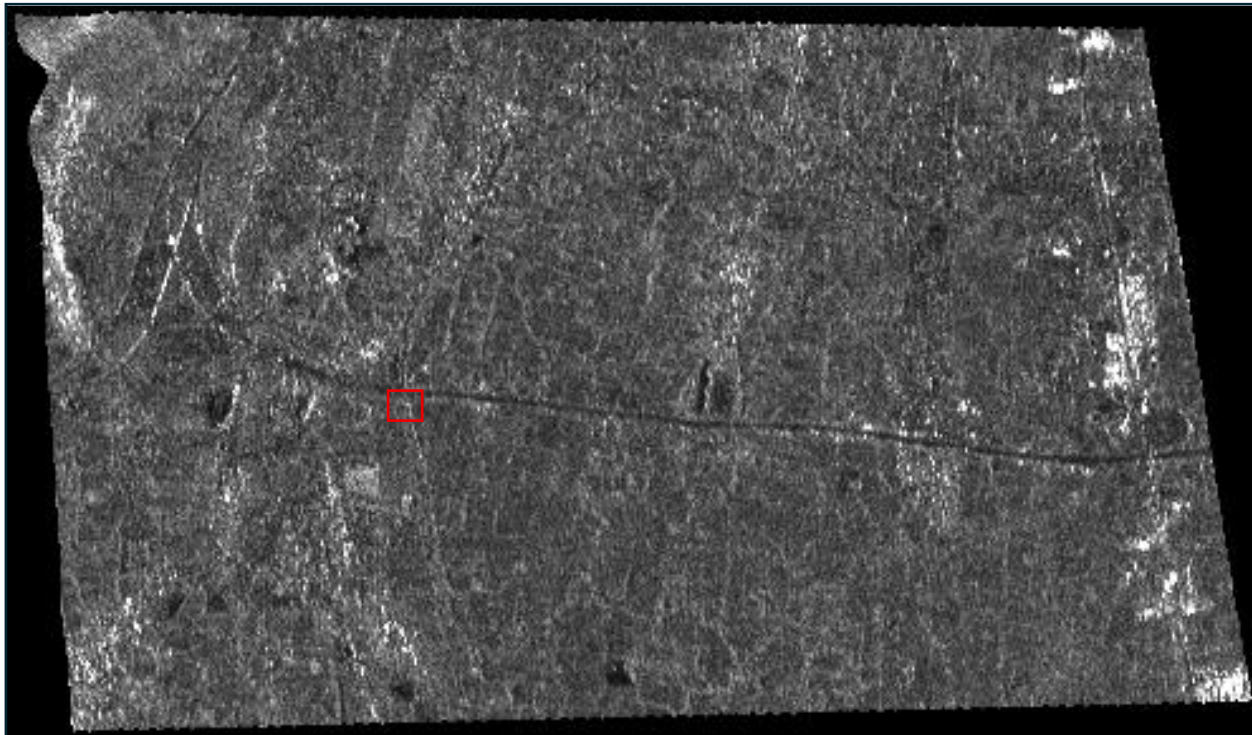
Absolute Orbit – 54706Frame – 519

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

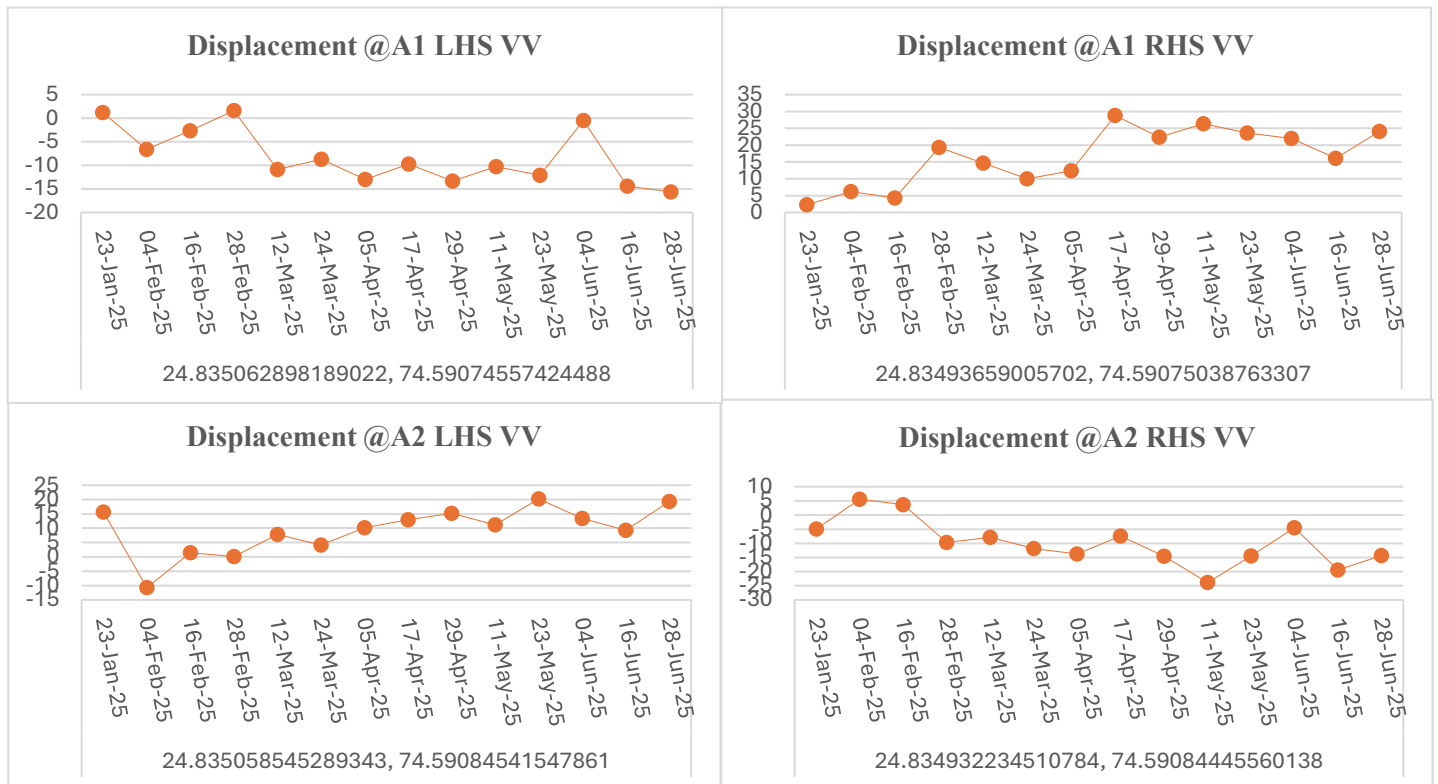
VUP – CH-893+270

Length – 12.17M

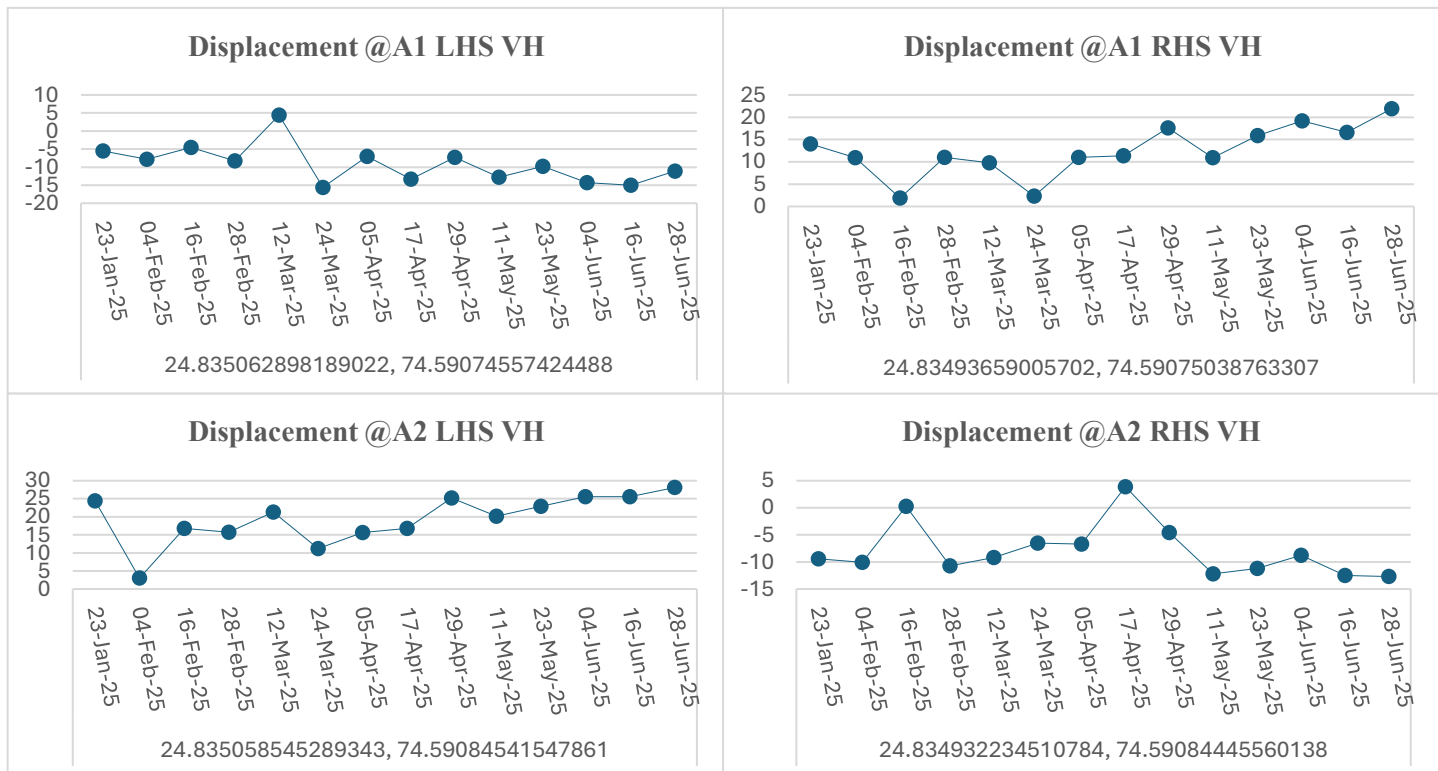


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



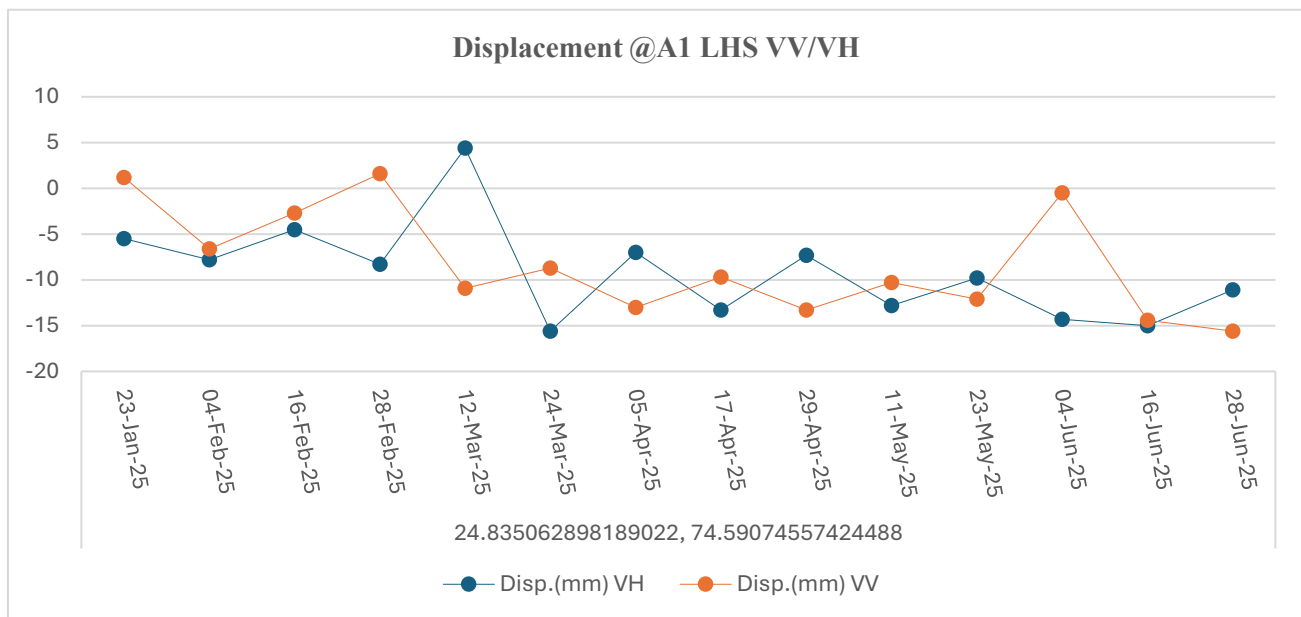
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.835062898189022, 74.59074557424488)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement values range approximately from –15 mm to +2 mm, indicating moderate vertical movement dominated by settlement phases with brief uplift events.
- **Horizontal Displacement (VH)** Horizontal displacement at A1 LHS remains predominantly negative, ranging approximately from –16 mm to +4 mm.

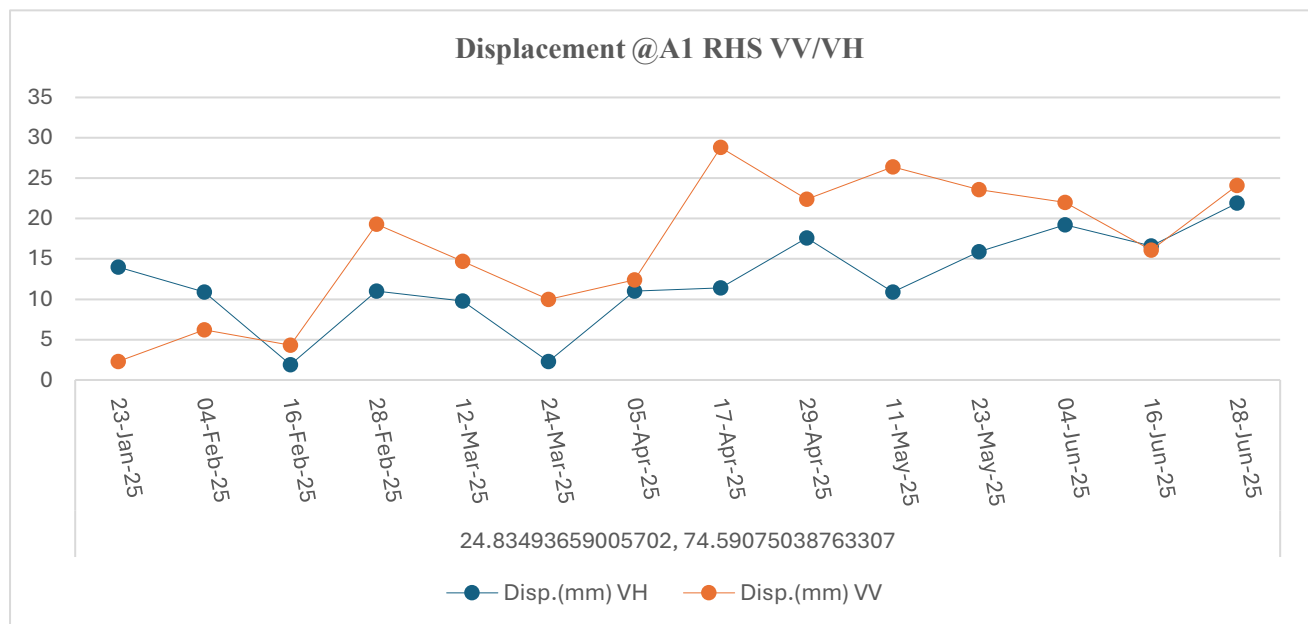
Critical Events & Anomalies:

- **February 2025:** VV shows a brief uplift (around +2 mm) while VH remains negative, indicating temporary vertical recovery without corresponding lateral improvement.
- **April 2025:** Both VH and VV remain consistently negative with moderate fluctuations, indicating a period of sustained settlement and lateral restraint.
- **June 2025:** Both VH and VV converge toward negative extremes (–14 to –15 mm), indicating continued settlement with persistent lateral displacement, pointing toward longer-term deformation trends.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.83493659005702, 74.59075038763307)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement at A1 RHS shows an overall positive trend, ranging approximately from +2 mm to +22 mm.
- **Vertical displacement (VV)** Vertical displacement values range approximately from +2 mm to +28 mm, indicating moderate to high vertical movement with pronounced uplift phases dominating over settlement.

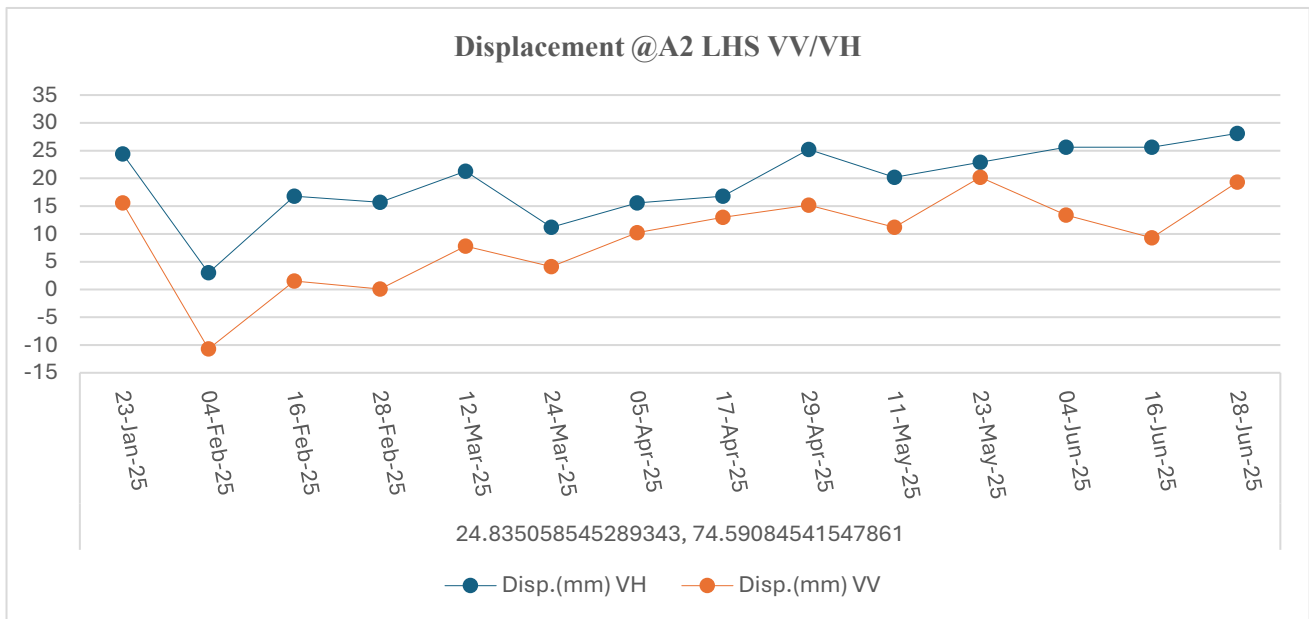
Critical Events & Anomalies:

- **February 2025:** VH reaches its minimum (+2 mm) while VV remains positive, suggesting a temporary reduction in lateral displacement without vertical settlement.
- **May 2025:** VH continues to increase steadily while VV shows a mild decline, indicating lateral progression with partial vertical stabilization.
- **June 2025:** VH and VV converge toward higher positive values (+16 to +24 mm), reflecting continued uplift accompanied by stable lateral displacement, consistent with long-term deformation behaviour.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.835058545289343, 74.59084541547861)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement at A2 LHS remains consistently positive, ranging approximately from +3 mm to +30 mm.
- **Vertical Displacement (VV)** Vertical displacement ranges approximately from –10 mm to +20 mm.

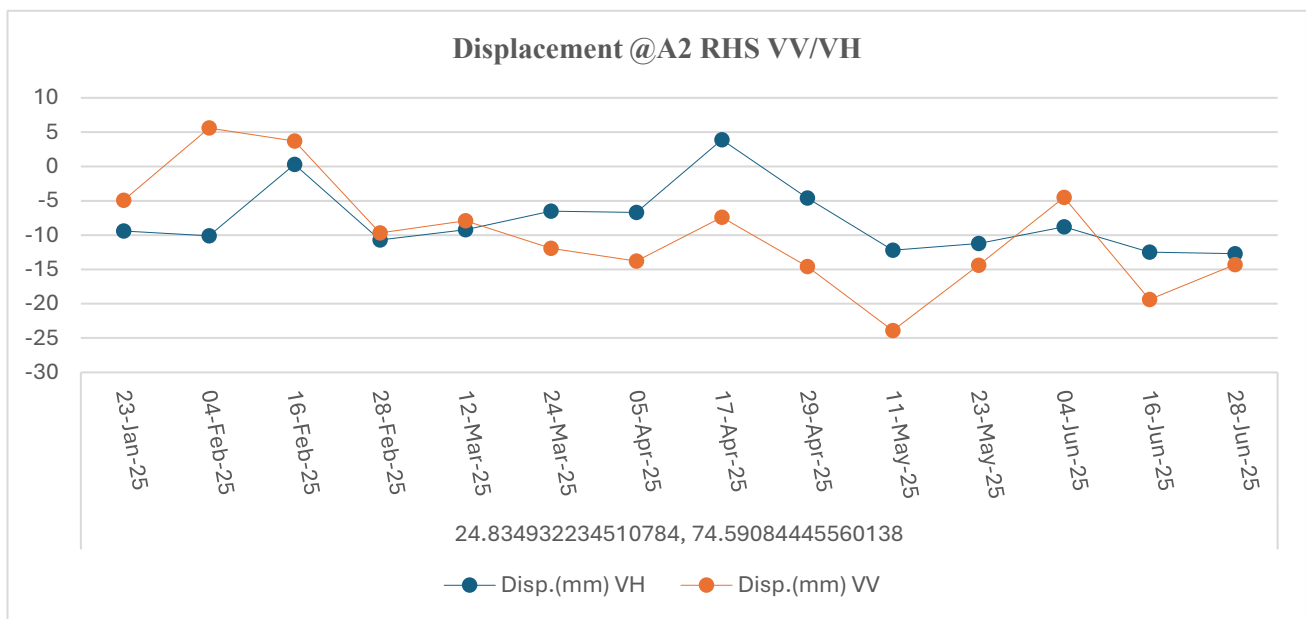
Critical Events & Anomalies:

- **February 2025:** VV records a significant negative peak (–10 mm) while VH drops to its minimum (+3 mm), indicating a short-term settlement and lateral relaxation event.
- **May 2025:** VH remains consistently high while VV fluctuates moderately, reflecting stable lateral deformation with controlled vertical variations
- **June 2025:** VH attains its maximum (+30 mm) and VV trends upward again (+18–20 mm), indicating continued long-term lateral movement accompanied by renewed uplift, consistent with progressive structural behaviour

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.834932234510784, 74.59084445560138)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement varies over a wider range, approximately from +5 mm to –25 mm, indicating pronounced vertical settlement with intermittent uplift phases.
- **Horizontal (VH) displacement** Horizontal displacement at A2 RHS remains predominantly negative, ranging approximately from –15 mm to +4 mm.

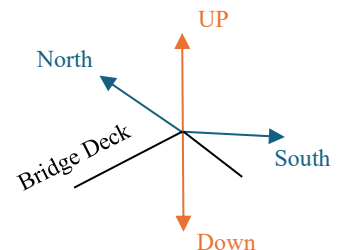
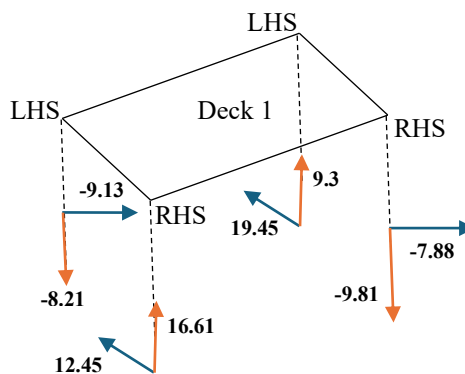
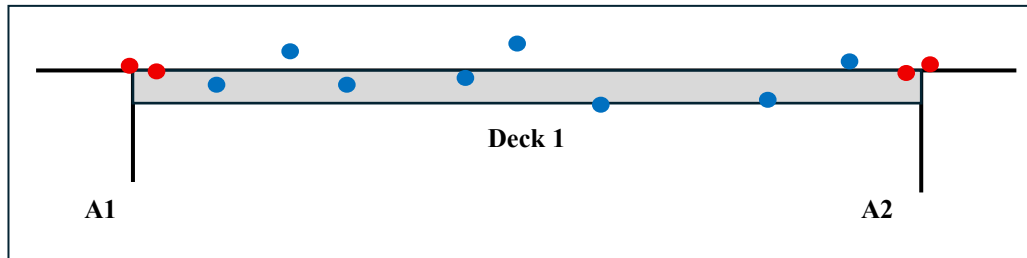
Critical Events & Anomalies:

- **February 2025:** VV shows a brief uplift peak (+5 mm) while VH approaches near-zero values, suggesting a temporary vertical recovery with reduced lateral restraint.
- **June 2025:** Both VH and VV remain negative, with VV again trending downward (~–20 mm), suggesting persistent long-term settlement with no sustained lateral recovery.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits distinct non-uniform vertical displacement, indicating differential behaviour across the deck width and support location. maximum uplift is observed at the central lower deck region, reaching approximately +16.6 mm, representing the dominant vertical response in this configuration.

Horizontal Displacement Behaviour:

- Horizontal displacement shows clear directional variability, reflecting asymmetric and torsional behaviour of the deck slab. The central deck region experiences the largest horizontal displacement, reaching approximately +19.5 mm, indicating strong lateral movement concentrated near the mid-deck.

CONCLUSION:

- The bridge deck slab demonstrates a highly non-symmetric deformation pattern, characterized by large lateral displacement at the central deck region and alternating uplift and settlement across LHS and RHS locations. Overall, the observed behaviour reflects complex interaction between vertical support response and lateral load redistribution, warranting continued monitoring and detailed correlation with loading and boundary conditions.